

ADENYLYL CYCLASE AND THE CONTROL OF CELL DIFFERENTIATION IN *Dictyostelium discoideum*

W. ROOS, D. MALCHOW and G. GERISCH

Biozentrum der Universität Basel, 4056 Basle, Switzerland

Accepted 2 June, 1977

Adenylyl cyclase is part of a biochemical network that controls cell differentiation in *Dictyostelium discoideum*. At a certain stage of development the enzyme is rhythmically activated, with periods of about 8 min. These oscillations are superimposed upon an increase of the basal activity extending over a period of hours. The basal activity remains low in a mutant blocked at an early stage of development. In strain Ax-2 two periods of strongly increasing basal activity have been found: the first from 2 to 4 h after the end of the growth phase, the other beginning at about 8 h.

Based on the periodic regulation of adenylyl cyclase, cyclic AMP is released into the extracellular space in the form of pulses. Application of cyclic-AMP pulses, but not its continuous influx, stimulates the increase of basal adenylyl cyclase activity. Two other constituents of the cyclic-AMP signal system, cyclic-AMP receptors and cell-surface phosphodiesterase, are similarly controlled. The principal importance of positive feedback loops in a network controlling cell differentiation is discussed.

In the present paper the differentiation of a uniform population of cells of one type into a uniform population of another cell type is described. After the end of their growth phase, cells of *Dictyostelium discoideum* differentiate into cells which have the capacity to aggregate. Because this process occurs in the single cell state rather than in a solid tissue, intercellular control factors are easily amenable to experimental variation. We show that the differentiation of the cells is controlled by a biochemical network in which extracellular cyclic AMP acts as part of a positive feedback loop. Other constituents of the control circuit are cyclic-AMP receptors (Malchow et al., 1974; Green et al., 1975; Henderson, 1975), cyclic-AMP phosphodiesterase at the cell surface (Malchow et al., 1972) and adenylyl cyclase within the cells (Farnham, 1975; Cutler et al., 1975).

About 3 h after the end of the growth phase, the adenylyl cyclase in cells of *D. discoideum* Ax-2 can be transiently activated by the application of small pulses of cyclic AMP to living cells (Roos et al., 1976). This result suggests a functional link between the cyclic-AMP receptors at the cell surface and adenylyl cyclase. Later during cell development cyclic-AMP pulses are formed autonomously based on the oscillation of adenylyl cyclase activity. During the pulses the enzyme activity reaches 10-fold its basal level (Roos et al., 1977).

In contrast to earlier results in which no developmental regulation was observed (Rossomando et al., 1972), Klein (1976) has found an increase of the adenylyl cyclase activity between 1 to 4 h after the end of growth phase by a factor of more than 45. This large increase suggests that not only the ability of the enzyme to switch periodically into the activated state is developmentally controlled, but also its basal activity. Therefore, taken together, the more recent observations suggest that periodic changes of the activity in the minute range are superimposed on a slower change of the basal activity. Experimental separation of these two contributions to total activity seems to be necessary in studies on the regulation of adenylyl cyclase in *D. discoideum*.

MATERIALS AND METHODS

Cell culture Cells were cultivated in shaken suspensions according to Malchow et al. (1972). For strains v-12/M2 and aggr 50-2-7, *E. coli* B/r was used as a nutrient source; for Ax-2 the nutrient medium of Watts et al. (1970) supplemented with 1.8% maltose. Cells were washed 3–4 × in 17 mM Soerensen phosphate buffer, pH 6.0, at various stages of growth, and resuspended at $1 \cdot 10^7$ cells per ml. The end of washing was taken as zero time of cell differentiation. Pulses of cyclic-AMP were applied by dropping a solution of $3 \cdot 10^{-5}$ M cyclic AMP into the shaken cell suspension; for continuous influx the tip of a capillary was inserted into the suspension. In both cases the cyclic-AMP solution was supplied from a syringe driven by a Perfusor (Braun, Melsungen).

Adenylyl cyclase assay The enzyme was assayed as described by Roos et al. (1976) using the analytical procedure of Salomon et al. (1974). All determinations were done in duplicates except for a few samples lost in Fig. 5. For basal activity measurements, cell suspensions were cooled in an ice bath, centrifuged, resuspended at $2 \cdot 10^8$ cells per ml in homogenization medium (25 mM Tris-HCl pH 7.5, 0.25 mM ATP, 1 mM EDTA, 10 mM dithiothreitol, 0.25 M sucrose), sonicated for 5 sec, incubated 10–15 min in the ice bath for deactivation as shown in Fig. 1 and assayed for 5 min. (No significant loss in the ice bath of basal activity was found in the homogenates within 2 h.) 100 μ l Assay mixture contained 0.2–0.5 mg homogenate protein determined according to Lowry, the ATP concentration was 0.5 mM.

For the experiment shown in Fig. 1 the cell suspension contained $1 \cdot 10^8$ cells per ml in 17 mM phosphate buffer pH 6.0. 150 μ l Samples of the suspension were mixed with 450 μ l homogenization medium of the same final concentration as above and immediately sonicated. The assay mixture (Roos et al., 1976) contained 0.13 mg homogenate protein in 100 μ l, and 0.18 mM ATP.

Contact-site A activity (Fig. 5) Cell agglutination was quantitated by measuring light scattering according to Beug et al. (1973) in 0.2-ml samples of 17 mM phosphate buffer, pH 6.0, containing $5 \cdot 10^6$ cells and 0.8 mg BSA per ml plus 8 mM EDTA. All measurements (E) were related to light scattering in a single cell suspension (E_0), prepared from an aliquot in which BSA was replaced by aggregation-blocking Fab.

Cyclic-AMP binding to cell-surface receptors (Fig. 6) Binding of 10 nM ^3H -cyclic AMP to intact cells was measured in the cold according to Malchow et al. (1974) using $5 \cdot 10^{-4}$ M cyclic GMP for blocking phosphodiesterase. Five and 20 sec after the addition of the nucleotides the cells were centrifuged and the label was determined in the supernatant. Between 5 and 20 sec binding increased only slightly; the 20-sec values were plotted.

Activity of cell-surface phosphodiesterase (Fig. 6) Hydrolysis of $50 \mu\text{M}$ ^3H -cyclic AMP was determined according to Malchow et al. (1974) in freshly washed cell suspensions containing inhibitor of the extracellular phosphodiesterase. At 23°C , assays were stopped either after 5 and 15 min (early stages of development) or after 3 and 10 min (later stages with high phosphodiesterase activities).

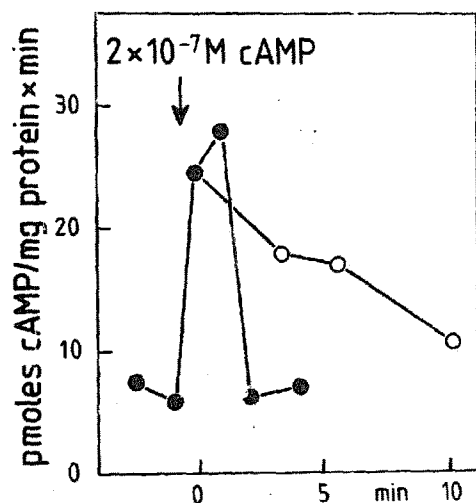


Fig. 1. Activation of adenylyl cyclase in cells stimulated by a cyclic-AMP pulse (●), and decline of the activated state after homogenization (○). At 4 h after the end of growth, a pulse of 200 nM cyclic AMP was administered to a cell suspension at 23°C . At the times indicated (●) samples of the suspension were sonicated for 5 sec and assayed within 10 sec for adenylyl cyclase. After 15 sec the assay was stopped. The homogenate prepared at 40 sec after stimulation and kept in an ice bath and assayed repeatedly to follow the decline of the activated state (○).

RESULTS

Activation by cyclic AMP, and basal activity of adenylyl cyclase

Fig. 1 shows the time course of adenylyl cyclase activation in cells stimulated by a cyclic-AMP pulse. A sonicate prepared during the activated state of the cyclase was incubated in an ice bath, and samples were assayed at various times. After 10 min in ice the activity had almost completely returned to basal activity. In the following experiments all homogenates were preincubated in the ice bath before the adenylyl cyclase was assayed. The

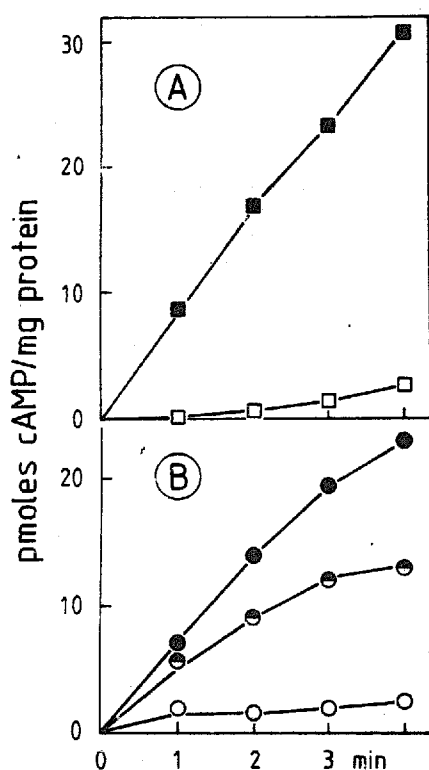


Fig. 2. Basal adenylyl cyclase activities during different stages of cell differentiation, and in homogenate mixtures from different stages. A: Wild-type strain v-12/M2 grown on bacteria. Homogenates were prepared either immediately after the end of growth (□), or 2.5 h later (■). Under the conditions used the activated state of adenylyl cyclase declines rapidly at 23°C in homogenates, so that after 1 min only basal activities are observed (Roos et al., 1976). In the curves shown here the rate of cyclic-AMP synthesis was not higher during the first minute than subsequently, indicating that the adenylyl cyclase was in the basal state when the assay was started. B: Strain Ax-2/206 grown axenically. Cells were harvested during exponential growth. Homogenates were prepared immediately after washing the cells free of nutrient medium (○), and 7 h later (●). Freshly prepared homogenates of the two stages were mixed after incubation for 10 min in ice and assayed immediately (◐).

enzyme activity was determined in the unfractionated homogenates. After centrifugation for 20 min at 30,000 *g* in 0.25 M sucrose, more than 80% of the activity was found in the pellet, indicating that most of the adenylyl cyclase had persisted during sonication in the membrane-bound state.

The adenylyl cyclase activities were much higher in aggregation-competent cells than in growth-phase cells, both in the bacteria-grown wild-type strain v-12/M2 and in the axenically grown strain Ax-2. The data shown in Fig. 2 demonstrate for both strains that these are differences of the basal activities of adenylyl cyclase. When homogenates of growth-phase and aggregation-competent cells were mixed, no loss of basal adenylyl cyclase activity was found. (The data shown in Fig. 2B do not exclude a slight activation.) This result renders it unlikely that the low activity found in growth-phase cells is due to the inactivation of adenylyl cyclase during the assay.

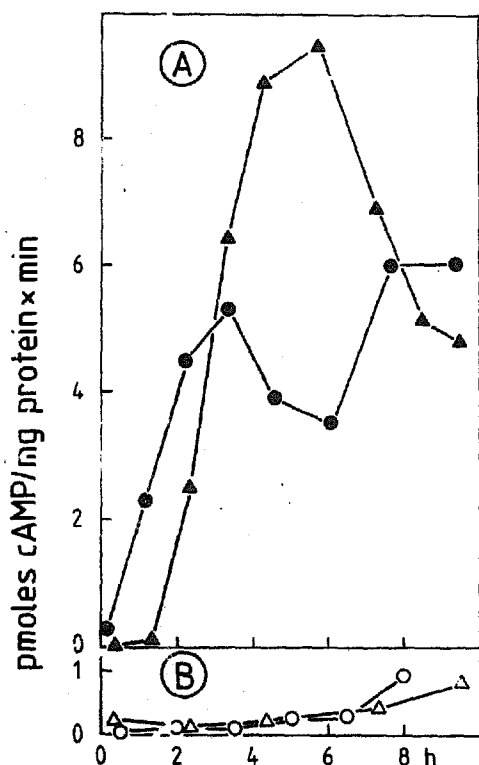


Fig. 3. Basal adenylyl cyclase activities during cell development in the wild-type v-12/M2 (A), and in its non-aggregating mutant aggr 50-2-7 (B). A: In the two experiments shown, the bacteria-grown cells were washed at different times during the growth phase. In cells washed at the end of exponential growth (about 4 h before the complete exhaustion of bacteria), the adenylyl cyclase began to increase after a delay of 1 h (▲). Only one peak of activity was obtained over a period of 10 h. Cells washed about 2 h later than before had obviously passed the lag phase before washing (●). Adenylyl cyclase activity increased immediately, and a second rise was observed after 6 h. B: Two experiments with aggr 50-2-7 in which the cells were washed free of bacteria at the end of exponential growth, about 5 h (○) and 4 h (△) before the complete exhaustion of bacteria.

Basal activities of adenylyl cyclase during cell differentiation

In *D. discoideum* v-12/M2, the activity of adenylyl cyclase increased by a factor of ≥ 20 from the end of growth up to the stage in which the cells were able to aggregate. As shown in Fig. 3A, a biphasic increase was observed under certain conditions. In aggr 50-2, a mutant blocked at an initial step of cell differentiation (Malchow et al., 1974), only a weak and delayed increase of the adenylyl cyclase activity was found (Fig. 3B).

During cell development of the axenically grown strain Ax-2, an initial rise of adenylyl cyclase activity ended with either a plateau or peak at 4 h after the end of growth, and was followed about 4 h later by a second increase (Fig. 4). As shown in Fig. 4B, the first increase of the adenylyl cyclase activity was reversibly blocked by cycloheximide, suggesting the involvement of de novo protein synthesis.

Control of basal activity of adenylyl cyclase activity by cyclic-AMP pulses

The beginning of autonomous oscillations during cell development is accelerated by the periodic application of cyclic-AMP pulses, not however by the continuous application of cyclic AMP (Gerisch et al., 1974). The following

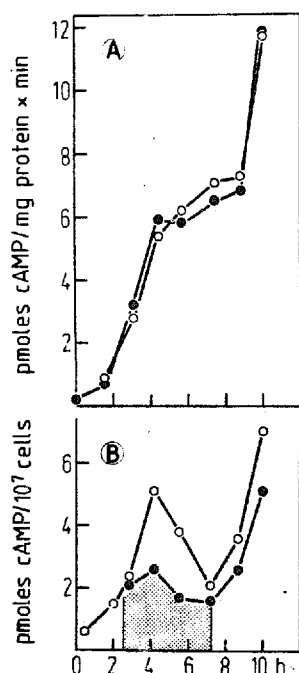


Fig. 4. Basal activity of adenylyl cyclase during development of axenically grown Ax-2 cells, and inhibition of its increase by cycloheximide. A: Cultures of clones 206 (●) and 214 (○), washed free of nutrient medium at the exponential growth phase. B: A culture of clone 206 in which two periods of increase were clearly resolved (○). Part of the cells were incubated with 250 μ g/ml cycloheximide (Sussman, 1966) at the beginning of the dotted zone, and washed at its end (●).

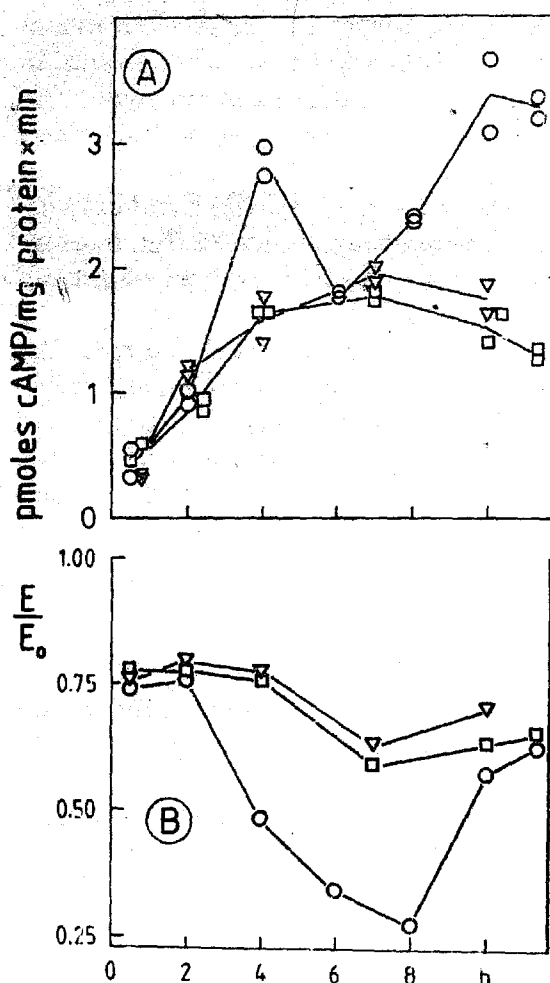


Fig. 5. Effect of cyclic-AMP pulses and of continuous cyclic-AMP influx on the regulation of basal adenylyl cyclase activity (A) and of contact sites A (B). Ax-2/206 cells were grown up to the late stationary phase before the growth medium was removed. Beginning immediately after washing, the cells were treated with pulses of cyclic AMP (○), with a continuous flux of cyclic AMP (□) or left without treatment (▽). Adenylyl cyclase assays were done in duplicates. Pulses and continuous flux were given at the same average rate. The pulse size was 5–14 nM, the period 5–6 min; the flux rate was 0.9–2.5 nmole per min × liter. (This variation was caused by the removal of samples from the cell suspension, so that the cyclic AMP applied per volume increased during the experiment.) Contact site A activity was determined according to Beug et al. (1973) by measuring cell agglutination in the presence of EDTA. Agglutination was assayed by recording light scattering: the absorption (E) of the sample relative to the absorption of a single cell suspension (E_0) is plotted on the ordinate. Low values indicate strong agglutination.

results show that this effect is correlated with an accelerated increase of the basal activity of adenylyl cyclase.

In axenically grown cells harvested at the late stationary phase, cell differentiation is inhibited (Gerisch et al., 1975a). Accordingly, the increase of adenylyl cyclase activity was less than in cells harvested at the exponential

growth phase. Using stationary phase cells, the effect of cyclic-AMP pulses on the basal activity of adenylyl cyclase was studied in three experiments. In all three, a biphasic increase with a similar time course as in normally differentiating cells was induced by cyclic-AMP pulses. The maximal activities obtained were however lower than during normal differentiation. Continuous application of cyclic AMP to the cells did not affect the basal activity of adenylyl cyclase. The results of one experiment are shown in Fig. 5, demonstrating the first basal activity peak at about 4 h after the end of growth and the second rise after about 8 h.

In two experiments the regulation of contact sites A was measured. The activity of these sites is reflected in the ability of the cells to form EDTA-stable contacts. Fig. 5 shows that the maximal activity of contact sites A was found in pulse-stimulated cells between the two peaks of adenylyl cyclase activity. In agreement with previous results (Gerisch et al., 1975), continuous influx of cyclic AMP had no significant effect on contact site A regulation.

Other constituents of the cyclic-AMP signal system

Both cyclic-AMP receptor sites and cell-surface phosphodiesterase are under developmental control (Malchow et al., 1974). As reported previously in short accounts (Gerisch et al., 1975, 1975b), the increase of both these constituents is stimulated by cyclic-AMP; and pulses are more efficient than continuous influx. Fig. 6 shows the stimulation in cells harvested at the late stationary phase. An accelerating effect of cyclic-AMP pulses was also

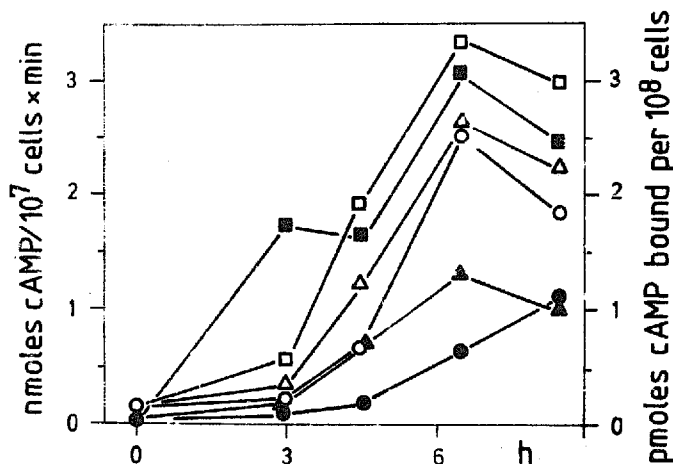


Fig. 6. Effect of cyclic-AMP pulses and of continuous cyclic-AMP influx on the regulation of cyclic-AMP receptors ($\square$, $\triangle$, $\circ$) and cell-surface phosphodiesterase ($\blacksquare$, $\blacktriangle$, $\bullet$). Ax-2 cells were harvested at the late stationary phase. Beginning 1 h after washing, the cells were either pulsed with cyclic AMP ($\square$, $\blacksquare$), subjected to a continuous influx of cyclic AMP ($\triangle$, $\blacktriangle$), or left without treatment ($\circ$, $\bullet$). The pulse size was 3–5 nM, the period 6 min. The continuous flux varied from 0.5 to 0.9 nmole cyclic AMP per min $\times$ liter, the same as the average rate of pulsatile application.

observed in normally differentiating cells harvested from the nutrient medium during exponential growth.

DISCUSSION

The results reported in this paper show a positive influence of cyclic-AMP pulses on the expression of three constituents of the cyclic-AMP signal system: adenylyl cyclase, cyclic-AMP receptors and cell-surface phosphodiesterase. Cyclic-AMP pulses represent the natural signal which is produced by *D. discoideum* cells during their differentiation (Gerisch and Wick, 1975). The possible reason why pulses are more efficient than continuous cyclic-AMP application has been discussed elsewhere (Gerisch et al., 1975c). It is believed that the optimal stimulus for the signal processing system in these cells is a change of the cyclic-AMP concentration in time, rather than a steady concentration. Such a behavior would be similar to signal perception in chemotactic bacteria (Brown et al., 1974; Spudich et al., 1975).

In the present paper attention is focused on the biochemical network involved in the differentiation of cells from the growth-phase to the aggregation-competent state. The three membrane constituents investigated are components of the network. They change from low to high activities while the network is functioning. A diagram of the network is shown in Fig. 7. The system can be described qualitatively as follows.

Cyclic-AMP pulses are formed in response to smaller extracellular pulses by receptor-mediated activation of adenylyl cyclase, and by transport of the synthesized cyclic AMP into the extracellular space (pathway A in Fig. 7). In response to reiterated pulses the receptor number increases and, in consequence, the response system becomes more sensitive (control loop B).

The increase of basal adenylyl cyclase activity indicates that the response

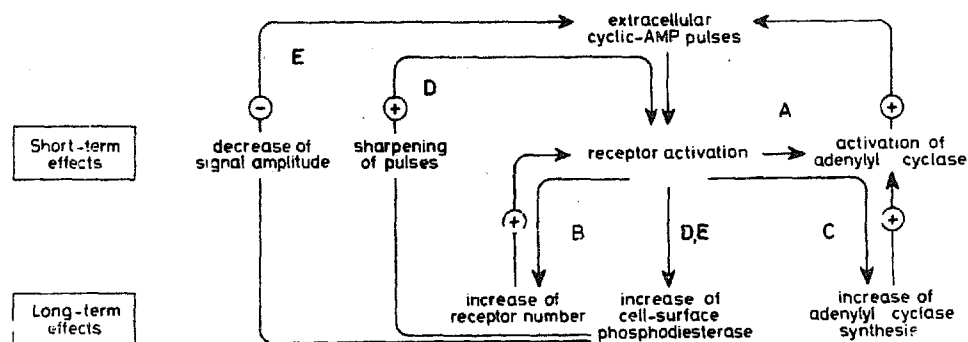


Fig. 7. Circuitry of a network thought to control cell differentiation in *D. discoideum*. Short-term controls imply the activity changes elicited by a single pulse of extracellular cyclic AMP, and the changes occurring during one period of autonomous oscillations. Long-term controls apply to changes within the hours required for cell differentiation to aggregation-competence; they probably involve de novo protein synthesis.

system becomes more efficient during cell development because of regulation not only on the receptor level, but also on the level of the enzyme (control loop C). In consequence, two positive feedback loops, B and C, lead to the increase during development of the amount of cyclic AMP formed in response to a given pulse, and of the amplitude of autonomous adenylyl cyclase oscillations. The increase of the response to a given pulse has been measured: it changes from non-detectability just after the end of growth up to a size of one to two orders of magnitude higher than the pulse applied for stimulation (Roos et al., 1975; Wick and Gerisch, in preparation).

The cell-surface phosphodiesterase has the function of removing the released cyclic AMP from one pulse to the next and thus to sharpen the pulses (Nanjundiah et al., 1976). This improves the efficiency of the extracellular signals because their pulsatile shape is important (control D). The phosphodiesterase exerts a negative effect on the signal amplitude and thus will act as an overshoot control, counteracting the positive interactions in the network (control E). These effects of the cell-surface phosphodiesterase are complemented by parallel regulation of extracellular phosphodiesterase (Klein, 1975).

The increase of the basal adenylyl cyclase activity is inhibited by cycloheximide, suggesting the involvement of de novo protein synthesis in the long-term control of the enzyme. Similar evidence has been obtained for the phosphodiesterase (Klein, 1975). The often biphasic increase of the basal adenylyl cyclase activity remains a subject of future investigations. Our studies have been done in cell suspensions where aggregate formation was inhibited by shaking. The biphasic increase might be another example of "quantal control" analogous to a new run of enzyme synthesis when a developmental sequence is repeated after the dissociation of slug cells (Sussman et al., 1972).

The principle exemplified by our results of *D. discoideum* is the transformation of cells into differentiated ones by the successive amplification of minor deviations from an initial state. One reason why the undifferentiated state is unstable in the absence of nutrients, is believed to reside in the positive interactions implemented in the network.

The importance of such instability phenomena in cell differentiation is not principally affected by the finding that in *D. discoideum* the first step of differentiation starts before the cyclic-AMP signal system begins to develop. During the first 2 h after the replacement of nutrient medium by a non-nutrient buffer, cyclic-AMP pulses have no effect on differentiation and other factors stimulate this first step of differentiation (Darmon et al., 1975; Gerisch et al., 1975, 1976).

The differentiated state is not only expressed in terms of high activities of the constituents of the cyclic-AMP signal system: contact sites A, for which a function in cell-to-cell adhesion has been suggested (Beug et al., 1973), are also regulated by cyclic-AMP pulses (Darmon et al., 1975; Gerisch et al., 1975a; and Fig. 5). This result indicates that the control network has outputs

to which other functional elements, not directly dealing with cyclic-AMP regulation, are connected.

The highest activity of contact sites A, marking the state of full aggregation-competence of the cells, did not coincide with either one peak of the basal adenylyl cyclase activity (Fig. 5). This result is in agreement with the idea that the adenylyl cyclase functions not only in the production of chemotactic signals during aggregation, but also in the control of cell differentiation.

ACKNOWLEDGEMENTS

The work was supported by the Schweizerischer Nationalfonds and the Deutsche Forschungsgemeinschaft.

REFERENCES

- Beug, H., F.E. Katz and G. Gerisch: *J. Cell Biol.* 56, 647–658 (1973).
 Brown, D.A. and H.C. Berg: *Proc. Natl. Acad. Sci. USA* 71, 1388–1392 (1974).
 Cutler, L.S. and E.F. Rossomando: *Exp. Cell Res.* 95, 79–87 (1975).
 Darmon, M., P. Erachet and L.H. Pereira da Silva: *Proc. Natl. Acad. Sci. USA* 72, 3163–3166 (1975).
 Farnham, C.J.M.: *Exp. Cell Res.* 91, 36–46 (1975).
 Gerisch, G. and B. Hess: *Proc. Natl. Acad. Sci. USA* 71, 2118–2122 (1974).
 Gerisch, G. and U. Wick: *Biochem. Biophys. Res. Commun.* 65, 364–370 (1975).
 Gerisch, G., H. Fromm, A. Huesgen and U. Wick: *Nature (London)* 255, 547–549 (1975a).
 Amsterdam) pp. 257–267 (1975).
 Gerisch, G. and U. Wick: *Biochem. Biophys. Res. Commun.* 65, 364–370 (1975).
 Gerisch, G., H. Fromm, A. Huesgen and U. Wick: *Nature (London)* 255, 547–549 (1975a).
 Gerisch, G., D. Malchow, A. Huesgen, V. Nanjundiah, W. Ross, U. Wick and D. Hulser: In: *Proceedings of the ICN-UCLA symposium on Developmental Biology*, eds. D. McMahon and C.F. Fox (W.A. Benjamin, Menlo Park, Calif.) (1975b).
 Gerisch, G., D. Hulser, D. Malchow and U. Wick: *Philos. Trans. R. Soc. London Ser. B* 272, 187–192 (1975c).
 Green, A.A. and P.C. Newell: *Cell* 6, 129–136 (1975).
 Henderson, E.J.: *J. Biol. Chem.* 250, 4730–4736 (1975).
 Klein, C.: *J. Biol. Chem.* 250, 7134–7138 (1975).
 Klein, C.: *FEBS Lett.* 68, 125–128 (1976).
 Malchow, D. and G. Gerisch: *Proc. Natl. Acad. Sci. USA* 71, 2423–2427 (1974).
 Malchow, D., B. Nagele, H. Schwarz and G. Gerisch: *Eur. J. Biochem.* 28, 136–142 (1972).
 Nanjundiah, V. and D. Malchow: *J. Cell Sci.* 22, 49–58 (1976).
 Roos, W. and G. Gerisch: *FEBS Lett.* 68, 170–172 (1976).
 Roos, W., C. Scheidegger and G. Gerisch: *Nature (London)* 266, 259–261 (1977).
 Ross, W., V. Nanjundiah, D. Malchow and G. Gerisch: *FEBS Lett.* 53, 139–142 (1975).
 Rossomando, E.F. and M. Sussman: *Biochem. Biophys. Res. Commun.* 47, 604–610 (1972).
 Salomon, Y. et al.: *Anal. Biochem.* 58, 541–548 (1974).
 Spudich, J.L. and D.E. Koshland, Jr.: *Proc. Natl. Acad. Sci. USA* 72, 710–713 (1975).
 Sussman, M.: *Proc. Natl. Acad. Sci. USA* 55, 813–818 (1966).
 Sussman, M. and P.C. Newell: In: *Molecular Genetics and Developmental Biology*, ed. M. Sussman (Prentice-Hall, Englewood Cliffs, N.J.) pp. 275–302 (1972).
 Watts, D.J. and J.M. Ashworth: *Biochem. J.* 119, 171–174 (1970).